Q1-Answer the following questions (10marks)

The determinant of hessian of the function $x^2 - 2y^2 - 4y + 6$ at point $(0, 0)$ is

Answer:

Given $f(4) = 3, f'(4) = 8, f''(4) = 11$, and that all other higher order derivatives of $f(x)$ are zero at $x = 4$, and assuming the function and all its derivatives exist and are continuous between $x = 4$ and $x = 5$, the value of $f(5)$ is

Answer:

The root of the equation $f(x) = 0$ is found by using **the Newton-Raphson method**. The initial estimate of the root is $x_0 = 3, f(3) = 5$. The angle the line tangent to the function $f(x)$ makes at $x = 3$ is $57°$ with respect to the x-axis. The next estimate of the root, x_1 most nearly is

Answer:

Given

$$[A] = \begin{bmatrix} 12.3 & -12.3 & 20.3 \\ 11.3 & -9 & -11.3 \\ 10.3 & -7 & 8.5 \end{bmatrix}, [B] = \begin{bmatrix} 2 & 4 \\ -5 & 6 \\ 12 & -20 \end{bmatrix}$$ then if $[C] = [A]\,[B]$, then C_{31}=

Answer:

The lower triangular matrix [L] in the [L][U] decomposition of matrix given below

$$\begin{bmatrix} 25 & 5 & 4 \\ 10 & 8 & 12 \\ 8 & 12 & 22 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ \ell_{21} & 1 & 0 \\ \ell_{31} & \ell_{32} & 1 \end{bmatrix} \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix}$$ is

Answer:

The following x, y data is given.

x	6	9	14
y	24	37	25

The Newton's divided difference second order polynomial for the data is given by $f_2(x) = b_0 + b_1(x - 6) + b_2(x - 6)(x - 9)$ The value of b_1 is most nearly

Answer:

The following

x	1	2	3	4
y	1	200	400	900

data is regressed with least squares regression to $y = a_0 + a_1 x$. The value of a_1 most nearly is

Answer:

The value of $\int_{0.4}^{2.0} e^x dx$ by using **4-segment** Simpson's **1/3** rule most nearly is

Answer:

The value of an integral $\int_a^b f(x)dx$ given using 1, 2, and 4 segments trapezoidal rule is given as **6.3460, 2.4408, and 1.5536**, respectively. The best estimate of the integral you can find using **Romberg integration** is most nearly

Answer:

Given $4\frac{dy}{dx} + 5y^2 = \sin x, \quad y(0.2) = 9$ and using a step size of $h = 0.3$, the value of $y(0.8)$ using Euler's method is most nearly

Answer:

Given $2\frac{dy}{dx} + y^2 = e^x, \quad y(0.6) = 8$, and using a step size of $h = 0.3$, the best estimate of $\frac{dy}{dx}(0.9)$ **Runge-Kutta 4th order method** is most nearly:

Answer:

Q2-Determine the minimum of the function $f(x,y) = x^2 + y^2$. Use point **(2, 1)** as the initial estimate of the optimal solution. Conduct one iteration. ***(12marks)***

Q3- Evaluate

$$I = \int_2^{2.6} \int_4^{4.9} \frac{1}{xy} dxdy$$

Take **dx = 0.2** in x-direction, **dy = 0.3** in y-direction. Use **Trapezoidal rule in x - direction** and **Simpson's 3/8 rule in y-direction.**

(10marks)

Q4-*Determine the values of* ***a*** *and* ***b*** *for which the system* ***(10marks)***

$$\begin{bmatrix} 3 & 0 & 0 \\ 9 & 3 & 2 \\ 1 & 1 & a \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} b \\ 3 \\ -1 \end{bmatrix}$$

1) has a unique solution 2) has no solution and 3) has infinitely many solutions

مثال: أوجد القيم العظمى والصغرى

$\Rightarrow$ Sol:- $x^2 - 2y^2 - 4y + 6$

$$\Rightarrow \frac{\partial f}{\partial x} = 2x \quad / \quad \frac{\partial f}{\partial y} = -4y - 4$$

$$\nabla f = \begin{bmatrix} 2x(0,0) \\ -4y - 4(0,0) \end{bmatrix} = \begin{bmatrix} 0 \\ -4 \end{bmatrix}$$

$$\Rightarrow \frac{\partial^2 f}{\partial x^2} = 2 \quad , \quad \frac{\partial^2 f}{\partial y} = -4 \quad / \quad \frac{\partial f}{\partial x \partial y} = \frac{\partial f}{\partial x \partial y} = 0$$

$$\Rightarrow H = \begin{bmatrix} 2 & 0 \\ 0 & -4 \end{bmatrix}$$

$$|H| = (2 \times -4) - (0 \times 0)$$

$$= -8$$

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⇒

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③ ⇒ solve ⇒ root → $f(x) = 0$

⇒ $x_0 = 3$ / $f(3) = 5$

⇒ $x = 3 \rightarrow 57^\circ \rightarrow \tan(57^\circ) =$ 0.484699

$f(3)$

⇒ $x_{i+1} = x_i - \frac{f(x_i)}{f(x_i)}$

$= 3 - \frac{f(3)}{f(3)} = 3 - \frac{5}{0.484699}$

$= -7.3456800$

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(4) $[C] = [A][B]$

$$\Rightarrow \begin{bmatrix} 12.3 & -12.3 & 20.3 \\ 11.3 & -9 & 11.3 \\ 10.3 & -7 & 8.5 \end{bmatrix} \cdot \begin{bmatrix} 2 & 4 \\ -5 & 6 \\ 12 & -20 \end{bmatrix} = \begin{bmatrix} 329.7 & -430.6 \\ -68 & 217.2 \\ 157.6 & -170.8 \end{bmatrix}$$

3×3 $\quad$ 2×2 $\quad$ $3*2$

$$C_{31} = 157.6$$

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(5) $[A] = [L] \cdot [U]$

$$\begin{bmatrix} 25 & 5 & 4 \\ 10 & 8 & 12 \\ 8 & 12 & 22 \end{bmatrix}$$

$$L_2 = \begin{bmatrix} 1 & 0 & 0 \\ l_{21} & 0 & 0 \\ l_{31} & l_{32} & 1 \end{bmatrix} \Rightarrow l_{21} = \frac{a_{21}}{a_{11}} = \frac{10}{25}$$

$$= 0.4$$

$$l_{31} = \frac{a_{31}}{a_{11}} = \frac{8}{25}$$

$$= 0.32$$

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$$\Rightarrow \begin{bmatrix} 25 & 5 & 4 \\ 10 & 8 & 12 \\ 8 & 12 & 22 \end{bmatrix} \Rightarrow -2R_1 + 5R_2$$

$$\begin{bmatrix} 25 & 5 & 4 \\ 0 & 30 & 52 \\ 8 & 12 & 22 \end{bmatrix} \Rightarrow -8R_1 + 25R_3$$

$$\begin{bmatrix} 25 & 5 & 4 \\ 0 & 30 & 52 \\ 0 & 260 & 518 \end{bmatrix} \Rightarrow \begin{bmatrix} a'_{11} & a'_{21} & a'_{31} \\ a'_{21} & a'_{22} & a'_{23} \\ a'_{31} & a'_{32} & a'_{33} \end{bmatrix}$$

$$l_{32} = \frac{a'_{32}}{a'_{22}} = \frac{260}{30} = 8.66667$$

$$L = \begin{bmatrix} 1 & 0 & 0 \\ 0.4 & 0 & 0 \\ 0.32 & 8.6667 & 1 \end{bmatrix}$$

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⇒ (6) ⇒ Sol:-

x	y	b_1	b
6	24		
9	37		
14	25		

$$\Rightarrow b_1 = \frac{f(x_1) - f(x_0)}{x_1 - x_0}$$

$$= \frac{37 - 24}{9 - 6} = 4.3333$$

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(7) ⇒ Sol:- $y = a_0 + a_1 x$

$$\begin{bmatrix} n & \Sigma x_i \\ \Sigma x_i & \Sigma x_i^2 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \end{bmatrix} = \begin{bmatrix} \Sigma y_i \\ \Sigma x_i y_i \end{bmatrix}$$

$$\begin{bmatrix} 4 & 10 \\ 10 & 30 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \end{bmatrix} = \begin{bmatrix} 150 \\ 520 \end{bmatrix}$$

$$\Rightarrow 4a_0 + 10a_1 = 150$$

$$10a_0 + 30a_1 = 520$$

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$\Rightarrow$ (8) $\int_{0.4}^{2.0} e^x \cdot dx$ / $n = 4$ / (1/3)

(نقسم ونقي وإذا أحكيوا) نقاط (5) $\Rightarrow$

$\Rightarrow$ 5.0 :- $n = 4$

$x_0 = 0.4$

$x_4 = 2$

$h = \frac{b-a}{n}$

$= \frac{2 - 0.4}{4}$

$= 0.4$

$x_0 = 0.4$

$x_1 = 0.8$

$x_2 = 1.2$

$x_3 = 1.6$

$x_4 = 2$

$$\int_a^b f(x) \cdot dx = \frac{h}{3}\left[f(x_0) + 2(f x_2) + 4(f(x_1) + f(x_3)) + f(x_4)\right]$$

$$= \frac{0.4}{3}\left[1.491825 + 2\left(\frac{3.320116}{}\right) + 4\left(\frac{2.225541}{} + 4.95303\right) + 7.389056\right]$$

$$5.898052$$

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$\Rightarrow$ (b) $\Rightarrow$ sol$_n$ $n = 1, n = 2, n = 4$

$n = 3$

$\Rightarrow$ درجات الحرية $\Rightarrow$ تدريبي (b) نقاط

$\Rightarrow$

(i, k)	2st	2nd
R(1,0)		
R(2,0)	R(2,1)	
R(3,0	R(3,1)	R(3,2)

$\Downarrow$

(i, k)	2st	2nd
6.3460		
2.4408	2.236533	
1.5536		

$\Leftarrow$ معادلة

$$R(2,1) = \frac{4^1 \times R(2,0) - R(1,0)}{4^1 - 1}$$

$$= \frac{4 \times 2.4408 - 6.3460}{3}$$

$$= 1.13906$$

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$$\Rightarrow R(3,1) = \frac{4^1 \times R(3,0) - R(2,0)}{4^1 - 1}$$

$$= \frac{4 \times 1.5536 - 2.4408}{3}$$

$$= 1.2578667$$

$$\Rightarrow R(3,2) = \frac{4^2 \times R(3,1) - R(2,1)}{4^2 - 1}$$

$$= \frac{16 \times 1.2528627 - 1.1390625}{16 - 1}$$

$$= 1.2657827$$

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(10) $\Rightarrow$ sol:- $4\frac{dy}{dx} + 5y^2 = \sin x$

$\Rightarrow \frac{dy}{dx} = \frac{\sin x - 5y^2}{4}$ / $y = 9$

$x = 0.2$

$h = 0.3$

Euler $\Rightarrow \sum y_{i+1} = y_i + f(x_i, y_i)h$

$\Rightarrow$ (2 steps) عندي خطوتين

$\Rightarrow$ [1] $\Rightarrow y(0.5) = y(0.2) + f(0.2, 9) * 0.3$

$= 9 + (-101.2 * 0.3)$

$= -21.36$

$\Rightarrow$ [2] $y(0.8) = y(0.5) + f(0.5, -21.36) * 0.3$

$= -21.36 + (-570.19 * 0.3)$

$= -192.417$

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$\Rightarrow$ (11) $\Rightarrow$ sol: RK(4)

$$\Rightarrow 2\frac{dy}{dx} + y^2 = e^x$$

$$\Rightarrow \frac{dy}{dx} = \frac{e^x - y^2}{2}$$

$y = 8$

$x = 0.6$

$h = 0.3$

2 steps

$$\Rightarrow k_1 = hf(x_i, y_i) = 0.3 * f(0.6, 8)$$

$= -9.326682$

$$\Rightarrow k_2 = hf(x_i + 0.5h, y_i + 0.5k_1)$$

$= 0.3 f(0.75, 3.336659)$

$=$

$=$

$$k_3 = hf(x_i + 0.5h, y_i + 0.5k_2)$$

$$= 0.3 f(0.75, -5.274592)$$

$$= -15.4493681$$

$$\Rightarrow k_4 = hf(x_i + h, y_i + k_3)$$

$$= 0.3 (0.9, -2.4493981)$$

$$= -26.51653025$$

$$\Rightarrow y(0.9) = 8 + \frac{1}{6}\{-9.326182 + 2(-4.5081466) + 2(-15.4493681) + -26.5163025\}$$

$$= -4.62632522$$

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